

SPL1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

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Project Title:
Short-Term Traffic Congestion Prediction and Alternate Route Recommendation System

Project Description
This project predicts short-term traffic congestion (15–30 minutes ahead) for selected road segments using time-series traffic data, and recommends a less congested alternate route when congestion is high. Raw sensor data is cleaned and converted into lag, rolling-average, and time-based features. Congestion is then predicted using three models implemented entirely from scratch: a moving-average baseline, a linear regression model trained with gradient descent, and a feedforward neural network trained with manual backpropagation. Model performance is compared using MAE and RMSE, and predictions are classified into congestion levels (Free-flow / Moderate / Heavy).

Objective

- Predict short-term traffic congestion (15–30 min ahead) for selected road segments
- Recommend a less congested alternate route between two points

Input:

- Recent traffic speed readings for a road (time-series) + time-based features

Output:

- Predicted traffic speed/congestion level 15–30 minutes ahead
- (Stretch) Recommended alternate route if congestion is high

Core Features

- **Data Cleaning & Feature Engineering** — lag values, rolling averages, time-of-day/day-of-week patterns
- **Congestion Prediction** — baseline, linear regression, and neural network models
- **Model Evaluation** — comparison using MAE and RMSE
- **Congestion Classification** — Free-flow / Moderate / Heavy
- **Alternate Route Recommendation** — Dijkstra's algorithm with congestion-weighted edges

Languages or Tools to Be Used:

- **Language:** C++
- **Libraries:** None — implemented from scratch (STL containers such as vector/string only)
- **OS:** Linux / Windows
- **Compiler:** GCC / G++

Machine Learning Models (from scratch):

- Simple Moving Average (baseline)
- Linear Regression (Batch Gradient Descent)
- Feedforward Neural Network / MLP (manual forward propagation and backpropagation)

Algorithms:

- Gradient Descent
- Backpropagation
- Moving Average
- Dijkstra's Algorithm (for route recommendation)

Supervisor's Name: Professor Dr. Naushin Nower

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory, and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____